Android, everyone has the Linux kernel in their hands. Android acts as a gateway for Linux to reach all kinds of people. The FOSS community believes in Android, but since Android poses a lot of problems due to the closed nature of its source code, some people thought of creating a mobile operating system without relying on any closed or proprietary code or services. That's how Replicant was born.

Most of Android's non-free code deals with hardware such as the camera, GPS, RIL (Radio interface layer), etc. So, Replicant attempts to build a fully functional Android operating system that relies completely on free and open source code.

The project began in 2010—named after the fictional Replicant androids in the movie 'Blade Runner'. Denis 'GNUtoo' Carikli and Paul Kocialkowski are the current lead developers for the Replicant.

In the beginning, they began by writing code for the HTC 'Dream' in order to make it a fully functional phone that did not rely on any non-free code. They made a little progress such as getting the audio to work with fully free and open source code, and after that they succeeded in making and receiving calls. You can find a video of Replicant working on the HTC Dream on YouTube.

The earlier versions of Replicant were based on AOSP (Android Open Source Project) but in order to support more devices, the base was changed to Cynogenmod—another custom ROM which is free but still has some proprietary drivers. The Replicant version 4.2 was released on January 22, 2014, which is based on Cynogenmod 10.1.

On January 3, 2014, the Replicant team released its full-libre Replicant SDK. You've probably noticed that the Android SDK is no longer open source software. When you try to download it, you will be presented with lengthy 'terms and conditions', clearly stating that you must agree to that license's terms or you are not allowed to use that SDK.

Replicant is all about freedom. As you can see, the Replicant team is labelling it the truly free version of Android. The team didn't focus much on open source, although the source code for Replicant is open to everyone. When it comes to freedom, from the users' perspective, the word simply means that they are given complete control over their device, even though they might not know what to do with that control. The Replicant team isn't making any compromises when it comes to the user's freedom. Although there may be some trade-offs concerning freedom, the biggest challenge for the Replicant team is to write hardware drivers and firmware that can support various devices. This is a difficult task since one Android device may differ from another. It's not surprising that they mainly differ in their hardware capabilities. That is why some apps that work well on one device may not necessarily work well on another. This problem could be solved if device manufacturers decide that the drivers and firmware should be given to the public, but we all know that's not going to happen. That's why there are some devices running on Replicant that still don't have 3D graphics, GPS, camera access, etc, but as mentioned earlier, people who value their freedom above all else, find Replicant very appealing.

The Replicant team is gradually making progress in adding support for more devices. For some devices, the conversion from closed source to open source becomes cumbersome, which is why these devices are rejected by the Replicant team.

F-Droid

One of the reasons for the grand success of Android is the wide range of apps that is readily available on the Google Play store for anyone to download.

For Replicant, you cannot use Google Play but you can use an alternative—F-Droid, which has only free and open source software.

The problem with Google Play is that many apps on it are closed source. So since we may not be able to look at their source code, there's a great possibility of an app that could spy on you or worse, steal your data being installed on it. By installing apps from Google Play, users inadvertently promote non-free software. Some apps also track their users' whereabouts.

F-Droid, on the other hand, makes sure all apps are built from their source code. When an application is submitted to F-Droid, it is in the form of source code. The F-Droid team builds it into a nice APK package from the source, so the user is assured that no other malicious code is added to that app since you can view the source code.

The F-Droid client app can be downloaded from the F-Droid website. This app is extremely handy for downloading and installing apps without hassle. You don't need an account but can install various versions of apps provided there. You can choose the one that works best for you and also easily get automatic updates.

If you're an Android user but want FOSS on your device, F-Droid is available to you. You have to allow your device to install apps from sources other than Google Play (which would be F-Droid). Using the single F-Droid client, you can easily browse through various sections of apps and easily remove the installed apps in your device or update your apps.

Using Replicant doesn't grant your device complete protection, but it can make your device less vulnerable to threats. It can offer you real control over your device and you can enjoy true freedom. If your device doesn't support Replicant, you can use Cynogenmod instead, which is officially prescribed as an alternative to Replicant.

As Benjamin Franklin put it, "Those who give up essential liberty to purchase a little temporary safety, deserve neither liberty nor safety." It's up to you to choose between liberty and temporary safety. END

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The author has completed a B.E. in computer science. As he is deeply interested in Linux, he spends most of his leisure time exploring open source.